NASA Space Apps Challenge 2026
Challenge: *Planet X and SPHEREx*
Local event: Milano, Politecnico di Milano, 14 and 15 November 2026

SPHEREx Blink

A blink viewer for the public SPHEREx images

Release v1.0.0, technical documentation

Team **Observationally Speaking**

Jessica Syafaq Muthmaina
Joowon Sohn

Release date 15 November 2026
Source code `https://github.com/yourastrophysicist/SPHEREx-site`
Live page `https://www.yourastro.space/SPHEREx/`

Abstract

SPHEREx Blink is a public web page that shows the same small patch of sky in every SPHEREx visit, one after the other, like a flipbook. On a fixed north-up grid the field stars stand still, so anything close enough to move, or anything that changes brightness, stands out to the eye. The page has two steps. A whole-sky map shows where the patch sits, and a blink viewer plays the SPHEREx images of it, in any of the six detectors, as one median image per six-month sky pass or as every single exposure. Gaia DR3 and CatWISE2020 sources are drawn on top, each moved to the date of the image with its own proper motion. The five test targets of the challenge ship as static files in all six detectors, 60 views in all, so the page works on plain static hosting. Any other position goes through a small Python server that asks the NASA/IPAC Infrared Science Archive (IRSA) live. Every setting lives in the page URL, so one link is one view. This document describes release v1.0.0: the data path, the files the page reads, the page itself, how to run it, how it was tested, and what it cannot do.

Contents

1 The challenge and our answer **2**

2 The data path **2**
2.1 Which exposures . . . 2
2.2 Moving the target to each date . . . 3
2.3 Cutouts . . . 3
2.4 Cleaning each exposure . . . 3
2.5 One grid for every frame . . . 4
2.6 Passes . . . 4
2.7 Catalogue overlays . . . 4

3 Static data **4**
3.1 The files . . . 4
3.2 Format of one view . . . 4
3.3 Built on GitHub . . . 5

4 The server **5**

5 The page **5**
5.1 Step 01, the whole sky . . . 5
5.2 Step 02, the blink viewer . . . 5

6 Running it **7**

7 Tests on the challenge targets **7**
7.1 What the data hold . . . 7
7.2 What each target shows . . . 7

8 What it can and cannot find **8**

9 Use of earlier work and of AI **9**

10 Credits and data use **9**

References **9**

1 The challenge and our answer

The challenge asked for a public-facing web tool. The user enters a right ascension and a declination, and the tool plays the SPHEREx images of that patch as a short video, so that moving objects are easy to spot. It named five test targets, Barnard's Star, Epsilon Indi B, WISE 0855−0714, RW Aurigae and R Draconis, and pointed to WiseView as a model.

SPHEREx (Spectro-Photometer for the History of the Universe, Epoch of Reionization and Ices Explorer) was launched on 12 March 2025. It maps the whole sky every six months with six detectors between 0.75 µm and 5.0 µm. Each detector sits behind a linear variable filter, so the wavelength changes across the array, and one star seen in two exposures is in general seen at two slightly different wavelengths. The pixels are large, about 6.15″. This makes SPHEREx a good survey for slow, bright, infrared movers such as nearby stars and brown dwarfs, and for variable stars, and a hard one for faint fast objects.

Our answer has four parts.

1. A **data path** in Python, one function call from coordinates to a list of aligned frames (Section 2).
2. **Static data**: the five targets exported in all six detectors as compact JSON, built on GitHub so the raw cutouts never fill a laptop (Section 3).
3. A small **server** that runs the data path live for any other position (Section 4).
4. The **page**, with a whole-sky map and a blink viewer, in English, Indonesian and Italian (Section 5).

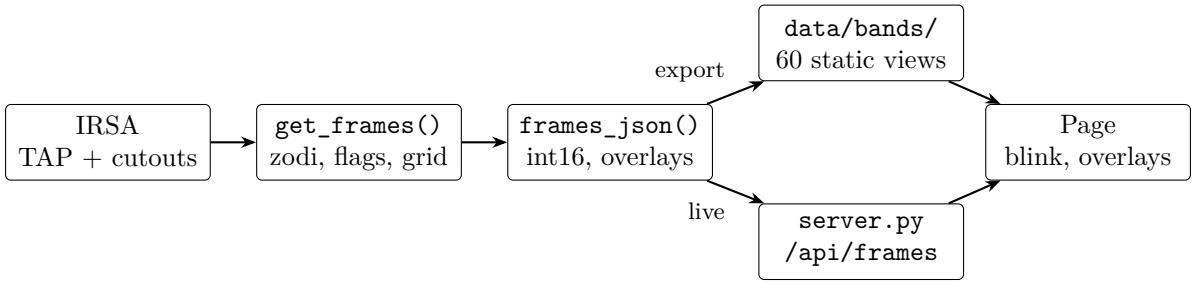


Figure 1: One code path, two ways to the page. The five targets come from static files, any other position from the live server.

2 The data path

All of this is in `tool/spherex_frames.py`. The public entry point is

```
frames, grid = get_frames(ra, dec, band="D1", fov=0.03, epoch=2016.0,
                          pmra=0.0, pmdec=0.0, name="target", mask=True,
                          progress=None, max_per_pass=None)
```

It returns the common grid as an `astropy` WCS and a list of frames, oldest first. Each frame holds the image on the grid, the variance, the date (MJD and decimal year), the wavelength at the target, the index of the six-month pass it belongs to, the predicted pixel of the target at that date, and the archive URI of the exposure.

2.1 Which exposures

The archive is queried through IRSA's TAP service with ADQL. A row is kept when the footprint polygon of the exposure contains the target and the detector matches:

```
SELECT a.uri, p.time_bounds_lower
FROM spherex.artifact a JOIN spherex.plane p ON a.planeid = p.planeid
WHERE 1 = CONTAINS(POINT('ICRS', ra, dec), p.poly)
  AND p.energy_bandpassname = 'SPHEREx-D1'
ORDER BY p.time_bounds_lower
```

The position used in the query is the target's position at 2026.0. The answer is cached on disk for 24 hours, because the query alone takes about 20 seconds and new SPHEREx data reach the archive about once a week. Fields near the ecliptic poles, such as R Draconis, are seen on almost every orbit and have hundreds of exposures, so at most 40 exposures per six-month pass are kept, spread evenly in time.

2.2 Moving the target to each date

Positions from Gaia DR3 are given at epoch J2016.0. A star with a large proper motion is many SPHEREx pixels away from that position by the time SPHEREx looks at it, so the position is moved to the date of each exposure. Over a few years a linear motion is enough:

$$\alpha(t) = \alpha_0 + \frac{\mu_{\alpha*}\,(t - t_0)}{\cos\delta_0}, \qquad \delta(t) = \delta_0 + \mu_\delta\,(t - t_0), \tag{1}$$

with (α_0, δ_0) at epoch t_0, $\mu_{\alpha*} = \mu_\alpha \cos\delta$ and μ_δ in $\mathrm{mas\,yr^{-1}}$, converted to degrees. Parallax is not applied. For Barnard's Star the parallax is $0.55\,''$, about a tenth of a pixel.

2.3 Cutouts

For each exposure a cutout centred on the moved position is asked from IRSA's image service. The cutout side is the field of view plus the distance the target travels over the whole time span plus two pixels,

$$s = \mathrm{fov} + \Delta t \sqrt{\mu_{\alpha*}^2 + \mu_\delta^2} + 2\,p, \qquad p = 6.15'', \tag{2}$$

so that the fixed grid is covered at every date even though the cutout centre moves. Downloads run in four parallel threads. The archive sometimes answers with a 503 or sends a file cut short, so each download is checked for the five extensions the tool needs and retried with an exponential back-off up to eight times. One exposure that still fails is skipped and logged instead of stopping the whole view.

Extension	What it holds	What we do with it
`IMAGE`	surface brightness in MJy/sr	the picture
`ZODI`	the pipeline's model of the zodiacal light	subtracted, so the sky level does not flicker between visits
`FLAGS`	one bit per problem (saturation, hot pixel, cosmic ray, and so on)	bits 0, 1, 6, 9, 10, 11, 15, 17, 19 count as bad
`VARIANCE`	noise per pixel	kept for light curves and centroid errors
`WCS-WAVE`	wavelength at each pixel	the wavelength at the target in each exposure

Table 1: The parts of each SPHEREx Level 2 file the tool uses.

2.4 Cleaning each exposure

The `ZODI` extension is subtracted from `IMAGE`. A pixel is flagged when any of the bits in the table above is set, the same set used in our prep script `coadd_spherex.py`, and with `mask` on flagged pixels become no data. The page shows unmasked pixels by default, because the saturated core of a bright star is flagged and a blink reads better when the star is still drawn. The spectral WCS gives the wavelength at the target's pixel, reported with every frame.

2.5 One grid for every frame

The grid is a gnomonic (TAN) projection, north up and east left, with SPHEREx-sized pixels. It is centred on the target's position at the mean date of all exposures and has an odd number of pixels per side, so the centre falls on a pixel. The page uses a field of $360''$, which gives 59×59 pixels. Because the grid is fixed on the sky and not on the target, the field stars stay still and a moving target walks across it. For every grid pixel the sky position is converted to the pixel coordinates of the exposure through the exposure's own WCS, and the image is sampled there with bilinear interpolation. The variance is sampled with nearest neighbour, so it is not smoothed.

2.6 Passes

SPHEREx covers a given point in visits a few days long, about six months apart. The frames are split into passes wherever two consecutive exposures are more than 60 days apart. Near the ecliptic poles the visits run together, so a polar field such as R Draconis falls into only two or three passes even with 80 or more exposures.

2.7 Catalogue overlays

Two catalogues come from the same IRSA TAP service, Gaia DR3 (positions at 2016.0) and CatWISE2020 (positions at MJD 57170, about 2015.4), which also holds objects too cold for Gaia, such as WISE 0855−0714. Each source is sent as its pixel on the grid at the catalogue epoch plus its motion in pixels per year, so the page can put it where it was on the date of any image. A star that walks in the data walks under its own circle.

3 Static data

The page is served from GitHub Pages, which cannot run Python. The five targets are therefore exported ahead of time through the same function the server uses (`frames_json`), so a static file and a live answer are byte for byte the same.

3.1 The files

- Each target in its best detector, written by `tool/export_static.py` to `data/<target>.json`, with a `_mask` version that blanks flagged pixels.
- Each target in all six detectors, 60 files in all, written by `tool/export_all_bands.py` to `data/bands/<target>_<band>.json`, with the same `_mask` versions.
- What each target is, in `data/targets.json`, why it is in the set, what to look for, and its numbers from Gaia DR3, AllWISE and CatWISE2020, checked on 14 November 2026.

3.2 Format of one view

One file is one target in one detector. It holds the grid (centre, size, reference pixel), the pixel scale, the detector, the catalogue overlays and the frames. Each frame carries its date, pass, wavelength at the target, predicted target pixel and file name at IRSA, so every image can be traced back to the archive. The image itself travels as little-endian int16 in base64,

$$v = \ell + s\,(c + 32767), \qquad s = \frac{v_{\max} - v_{\min}}{65534}, \tag{3}$$

with ℓ the minimum of the frame, c the stored code and $c = -32768$ meaning no data. This is about a third of the size of plain JSON numbers. The rounding error is 1/65534 of the range of the frame, far below the noise. A view is 0.4 to 0.8 MB.

3.3 Built on GitHub

The raw cutouts for all six detectors of the five targets come to several gigabytes, and on 14 November they filled the disk of the laptop we were working on. The export now runs on GitHub instead, through the workflow `.github/workflows/export-bands.yml`. Each target runs as its own job, the cutouts of each detector are deleted as soon as its JSON is written, and files that already exist are skipped, so a stopped run picks up where it left off. The workflow never commits. The finished files come back as one zip per target, which a team member checks and commits by hand.

4 The server

`tool/server.py` uses the Python standard library only, besides the packages the data path needs (`numpy`, `scipy`, `astropy`). It serves the page and two endpoints.

```
GET /api/frames?ra=..&dec=..&size=..&band=..&epoch=..&pmra=..&pmdec=..&mask=..&job=..
GET /api/progress?job=..
```

Inputs are checked ($0 \le \alpha < 360$, $-90 \le \delta \le 90$, field side between 20 and 900 arcsec, detector D1 to D6) and a bad input returns HTTP 400 with the reason. An archive failure returns HTTP 502 with the error text, so the page can tell the user what went wrong. While a new field downloads, `/api/progress` reports how many cutouts are done, and the page shows a progress bar. Results are kept in memory per view, and when the same view is asked twice at once the second request waits for the first download instead of starting its own.

5 The page

The page is plain HTML, CSS and JavaScript (`index.html`, `style.css`, `app.js`) in the design of yourastro.space, with a light and a dark theme, in English, Indonesian and Italian.

5.1 Step 01, the whole sky

One Aladin Lite map (CDS, Strasbourg) shows the whole sky, laid out after Brice Ménard's ultraviolet sky map. It has three shapes (a Mollweide map with the Milky Way across the middle, a turning globe seen from outside, and the view from inside) and a switch between the sky and the Earth, where NASA's Blue Marble and Black Marble show the planet SPHEREx circles. The sky can show SPHEREx DR1, the mission's all-sky release in 102 colours, with D1 to D6 buttons and a wavelength slider, or 2MASS, AllWISE and DSS2 for comparison. The five targets are markers that load into the viewer below, and the footprint of the field on screen is drawn on the map. A crosshair, a draggable lens that hands its centre to the viewer, and a fixed Home view come from our teammate's Time Lens prototype.

5.2 Step 02, the blink viewer

The viewer draws the frames on a canvas, pixel for pixel, with no smoothing.

Sky pass or exposure. With one image per *sky pass*, the default, each pass is reduced to its per-pixel median, and the blink steps through the passes. This removes most cosmic rays and single-exposure artefacts, and it matches the six-month rhythm in which a proper motion becomes visible. With one image per *exposure*, every exposure is shown on its own. The counter always gives both numbers, for example `pass 2 / 3 ů 60 exp` in the first mode and `exp 12 / 60 ů pass 1 / 3` in the second, so it is clear how much data stands behind the few pass images.

Detector. Buttons D1 to D6, or the keys 1 to 6, switch the detector. For the five targets this loads the matching file from `data/bands/`. The target's best detector is underlined, and choosing a target opens it on that detector.

Blink or difference. The difference view subtracts from each image the per-pixel median M of all images, $D_i = I_i - M$. Whatever never changes turns mid grey. Something that moved leaves a bright spot where it is and a dark spot where it was.

Stretch. Each image is stretched on its own, so the change of brightness with wavelength between exposures does not swamp the blink. The sky level m is the median of the finite pixels and the noise is the scaled median absolute deviation, $\sigma = 1.4826\,\mathrm{MAD}$. In the default `asinh` stretch a pixel value v becomes

$$t = \frac{\operatorname{arsinh}\left((v-\ell)/\beta\right)}{\operatorname{arsinh}\left((h-\ell)/\beta\right)}, \qquad \ell = m - 2\sigma, \quad \beta = 3\sigma, \tag{4}$$

clipped to $[0, 1]$, where h is the upper percentile set by the `contrast` setting. In the difference view the stretch is symmetric about zero, $t = \frac{1}{2} + \frac{1}{2}\operatorname{arsinh}(D/\beta)/\operatorname{arsinh}(T/\beta)$ with $\beta = 2\sigma$ and T the upper percentile of $|D|$. A linear stretch is also available.

Overlays, marker and orientation. Gaia DR3 and CatWISE2020 sources are drawn as circles down to a chosen magnitude, each at its place on the date of the image. A marker of four short ticks shows where the target should be, with a gap in the middle so it never covers the object. A compass and a $30''$ scale bar show orientation and size.

This target. Under the target list a panel says what to look for in the blink and gives the target's distance, motion in arcseconds and pixels per year, and best detector, read from `data/targets.json`. The table at the bottom of the page gives the full reason each target is in the set.

The URL is the state. As in WiseView, every setting is a URL parameter, and the URL is rewritten when a control changes. Table 2 lists the viewer's parameters. The map adds `shape`, `world`, `layer`, `frame`, `earth` and `cross`. Both the map and the viewer have a full screen view.

Parameter	Meaning	Default
`target`	preset: `Barnard`, `EpsIndB`, `WISE0855`, `RWAur`, `RDra`	`Barnard`
`ra`, `dec`	position in degrees at `epoch`	
`pmra`, `pmdec`	proper motion in $\mathrm{mas\,yr^{-1}}$	0
`epoch`	epoch of the position, decimal year	2016.0
`size`	field side in arcsec, up to 900	360
`band`	SPHEREx detector, D1 to D6	best for the target
`view`	`blink` or `diff`	`blink`
`mode`	`pass` (median of each pass) or `frames` (every exposure)	`pass`
`speed`	blink interval in ms	500
`stretch`	`asinh` or `linear`	`asinh`
`contrast`	upper percentile of the stretch	99.5
`invert`, `marker`, `mask`	1 or 0	0, 1, 0
`gaia`, `catwise`	catalogue overlay on or off	1, 0
`gmag`, `wmag`	faintest Gaia G and CatWISE $W2$ drawn	16, 16

Table 2: Viewer URL parameters. Keys: space plays or pauses, the arrows step, 1 to 6 choose the detector, `d` difference, `m` marker, `g` Gaia, `f` full screen.

6 Running it

The live page needs nothing installed. To look at a position other than the five targets, run the server on your own computer:

```
git clone https://github.com/yourastrophysicist/SPHEREx-site
cd SPHEREx-site
conda activate astrolab2 # any Python 3.10+ with numpy, scipy, astropy
python tool/server.py # then open http://localhost:8000/?target=Barnard
python tool/export_static.py # rebuilds data/ after a change to the data path
```

The first view of a new field asks IRSA for every exposure and downloads the cutouts. This takes a few minutes, and longer when the archive is busy. The files are then kept in `~/spherex_data` (or the folder in `$SPHEREX_DATA`) and the same view opens in seconds. To rebuild `data/bands/`, run the workflow *Export all detectors* from the Actions tab of the repository and commit the files it returns.

7 Tests on the challenge targets

Table 3 gives the inputs of the five presets. Positions and motions are from Gaia DR3 at J2016.0, except WISE 0855−0714, which is too faint for Gaia and uses the AllWISE position with the motion of Kirkpatrick et al. (2021). The last column is the detector where the target is easiest to see, which the page opens first.

Target	α (deg)	δ (deg)	epoch	$\mu_{\alpha*}$ (mas yr^{-1})	μ_δ	best
Barnard's Star	269.448503	4.739420	2016.0	−801.6	10362.4	D1
Epsilon Indi B	331.076453	−56.793812	2016.0	3982.0	−2466.8	D5
WISE 0855−0714	133.794760	−7.245146	2010.56	−8123.7	673.2	D6
RW Aurigae	76.956081	30.401205	2016.0	2.6	−27.4	D1
R Draconis	248.167378	66.755007	2016.0	−20.6	6.1	D3

Table 3: The five presets.

7.1 What the data hold

Table 4 gives the number of exposures and sky passes in each of the 60 views, as exported on 14 November 2026. Every view has data, and in every view the wavelength at the target falls inside the range of its detector.

Target	D1	D2	D3	D4	D5	D6
Barnard's Star	60 / 3	61 / 3	59 / 3	59 / 3	61 / 3	58 / 3
Epsilon Indi B	46 / 3	93 / 4	96 / 4	46 / 3	84 / 4	93 / 4
WISE 0855−0714	36 / 3	48 / 3	45 / 3	36 / 3	47 / 3	45 / 3
RW Aurigae	56 / 3	55 / 3	56 / 3	56 / 3	56 / 3	56 / 3
R Draconis	80 / 2	89 / 3	80 / 2	78 / 2	89 / 3	80 / 2

Table 4: Exposures / sky passes per target and detector. R Draconis has at most 40 exposures per pass kept (Section 2).

7.2 What each target shows

Barnard's Star, D1. 60 exposures over 3 passes. Across them the star lands on average within a tenth of a pixel of the position predicted by the Gaia proper motion, which checks the alignment

and the motion correction together. It walks north by about 10.4″ per year, which is 0.85 px per six-month pass, plain to see over the three passes. Its core is saturated and flagged, so with the mask on it shows as a moving hole.

Epsilon Indi B, D5. The pair of brown dwarfs is one blob at SPHEREx resolution and moves about 0.76 px per year, so in the difference view it shows as a dark and a bright spot side by side. Gaia gives a RUWE of 4.3, as expected for an unresolved pair.

WISE 0855−0714, D6. The coldest known brown dwarf is faint at SPHEREx's longest wavelengths and missing or very weak in D1 to D4. Stepping through the detectors shows it appear, which is why the tool keeps each detector apart. In CatWISE2020 the object is split into several entries with motions near 8″ per year, because it moved so far across the WISE co-adds.

RW Aurigae and R Draconis. These two barely move (0.004 and 0.003 pixels per year). They are in the set to show the other half of the job, a star that stays in place while its brightness changes. R Draconis is a Mira variable with a period of 246 days, and since six months is not a whole number of periods, each pass catches it at a different phase. For R Draconis the CatWISE overlay is empty, because IRSA returned no CatWISE2020 sources in that field. The Gaia overlay works there.

8 What it can and cannot find

Published forecasts put a single SPHEREx exposure at roughly $V \approx 17$ for a point source at 5σ per channel, and the all-sky map after four passes at about AB 19.5 to 19.9 (deeper near the ecliptic poles). These are forecasts, not our measurements. The blink works on single passes, so its practical limit is close to the depth of one pass.

Behind the challenge sits the question of a distant "Planet X". For a planet seen in reflected sunlight at opposition the flux falls as r^{-4},

$$m = m_\odot - 2.5 \log_{10} \frac{A_g R_p^2}{r^2 \Delta^2}, \qquad m_\odot = -26.74 \ (V), \tag{5}$$

with the radius R_p and the distances r and $\Delta \approx r$ in AU and A_g the geometric albedo. A planet of three Earth radii with $A_g = 0.3$ is at $V \approx 18.8$ at 300 AU and $V \approx 21.0$ at 500 AU. That is fainter than a single SPHEREx exposure, so this tool will not find a Planet Nine of the usual size and distance, and we say so on the page. What it does well is what SPHEREx is suited for, nearby stars and brown dwarfs with large proper motions and variable stars, in the infrared and on the whole sky.

Known limits of v1.0.0.

- Linear proper motion only, no parallax.
- One detector at a time. The wavelength changes slightly between exposures because of the linear variable filter. It is reported, not corrected.
- On the live site only the five targets open. Any other position needs the server running on the user's own computer, and its first view is slow because the cutouts come from IRSA live.
- Near the ecliptic poles the 60-day rule merges visits into few passes.
- The server is meant to run locally. It has no rate limiting.

9 Use of earlier work and of AI

Earlier work. The data path grows out of two prep scripts and a notebook made by Jessica Syafaq Muthmaina on 10 November 2026, before the event, to check that the SPHEREx data could be reached. The TAP query, the proper-motion centring, the retrying download and the flag mask come from there. The Milano Local Lead confirmed on 14 November 2026 that this earlier work may be used. It is kept apart in the `prep/` folder of the repository, with its own README and guide. The lens, the crosshair and the fixed Home of the whole-sky map come from Joowon Sohn's Time Lens prototype.

AI. Parts of the code and of this document were written with help from Claude (Anthropic). The team chose the method, checked the results against the archive and the expected motions, and is responsible for the content.

10 Credits and data use

- **SPHEREx** Level 2 spectral images and the DR1 all-sky cube: NASA/JPL-Caltech/IPAC, through the NASA/IPAC Infrared Science Archive (IRSA).
- **Gaia** DR3: European Space Agency, mission Gaia, processed by the Gaia Data Processing and Analysis Consortium (DPAC).
- **AllWISE** and **CatWISE2020**: NASA/JPL-Caltech, UCLA. Motion of WISE 0855−0714 from Kirkpatrick et al. (2021).
- **AAVSO VSX** for the period and range of R Draconis.
- **Aladin Lite** (CDS, Strasbourg), with 2MASS and AllWISE through IRSA and DSS2 through ESA.
- **Blue Marble** and **Black Marble**: NASA, through NASA GIBS.
- **WiseView** (Caselden et al.), the model for the blink, and Brice Ménard's ultraviolet sky map, the model for the whole-sky layout.
- Software: NumPy, SciPy, Astropy.

NASA does not endorse this project. Data are used under the terms of each archive.

References

[1] SPHEREx mission, NASA/JPL-Caltech/IPAC, `https://spherex.caltech.edu`.

[2] NASA/IPAC Infrared Science Archive, SPHEREx data and tutorials, `https://irsa.ipac.caltech.edu` and `https://caltech-ipac.github.io/irsa-tutorials`.

[3] Gaia Collaboration, Vallenari, A. et al. 2023, *Gaia Data Release 3: Summary of the content and survey properties*, A&A 674, A1.

[4] Marocco, F. et al. 2021, *The CatWISE2020 Catalog*, ApJS 253, 8.

[5] Kirkpatrick, J. D. et al. 2021, *The field substellar mass function based on the full-sky 20 pc census of 525 L, T, and Y dwarfs*, ApJS 253, 7.

[6] Caselden, D. et al. 2018, *WiseView: Visualizing motion and variability of faint WISE sources*, Astrophysics Source Code Library, ascl:1806.004.

[7] Bonnarel, F. et al. 2000, *The ALADIN interactive sky atlas*, A&AS 143, 33.

[8] Ménard, B., ultraviolet sky map, `https://menard.pha.jhu.edu/uvmap`.

[9] Astropy Collaboration 2022, *The Astropy Project: Sustaining and growing a community-oriented open-source project and the latest major release (v5.0)*, ApJ 935, 167.

[10] Earlier work: `prep/` in `https://github.com/yourastrophysicist/SPHEREx-site`, 10 November 2026.